

A 340- μ W TinyML Using LUT-Based Reservoir Computing on Low-Cost FPGAs

Kanta Yoshioka^{1b} and Hakaru Tamukoh^{1b}

Abstract—We propose a reservoir computing (RC) system that operates under extremely strict power consumption constraints of less than 340 μ W, which enables continuous operation for one year on a LR6 battery. By combining a look-up tables networks based RC (LUTNet-RC) with iCE40 series field-programmable gate arrays (FPGAs), the proposed system achieves high computational accuracy while significantly reducing power consumption compared with conventional TinyML devices. The proposed method achieves 93.1%, 98.6%, and 92.7% accuracy in real TinyML applications such as human activity recognition, epilepsy detection, and electrocardiogram signal analysis, respectively, while operating at about 254 to 335 μ W power consumption. This work shows that the proposed LUTNet-RC on iCE40 FPGAs are promising solutions for long-term operational machine learning application implementations in battery-powered edge devices.

Index Terms—Field programmable gate array, Internet of Thing (IoT), machine learning, neuromorphic computing, reservoir computing (RC), TinyML.

I. INTRODUCTION

THE DEMAND for edge machine learning (TinyML) is rapidly increasing owing to the proliferation of Internet of Thing (IoT) devices and wearable sensors [1]. For these TinyML devices, power consumption constraints are a major challenge. In particular, the total system power consumption must be kept below 340 μ W to achieve continuous operation for more than one year on a standard LR6 battery (3000 mWh). These strict power consumption constraints are at the level required by TinyML, which operates embedded in people and factory machinery in the healthcare and anomaly detection fields. In addition, low-cost manufacturing is crucial in TinyML deployments within embedded devices. Then, combining the low manufacturing costs and strict power constraints while ensuring optimal performance is challenging.

Reservoir computing (RC) is attracting attention for real-time data processing in edge devices owing to its low

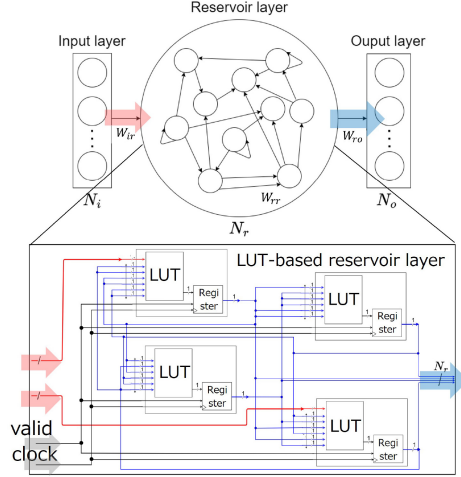


Fig. 1. RC model and LUT-based reservoir layer.

training, inferring cost, and high accuracy comparable to conventional deep-learning-based recurrent neural networks (DL-based RNNs) [2]. RC has been applied to various edge systems such as healthcare and anomaly detection, which require fitting into a variety of unique environments. However, conventional RC implementations, even with microcontroller units (MCUs) and field-programmable gate arrays (FPGAs), require power consumption of several mW to several tens of mW, making it difficult to satisfy the strict constraint of 340 μ W [2], [3].

In this work, we addressed the above limitations by focusing on look-up tables networks based RC (LUTNet-RC) [4], an RC model specialized for FPGA implementation, as shown in Fig. 1. In LUTNet-RC, the reservoir layer, which comprises most of the RC calculation, is realized using LUTNet quantization method. Consequently, the reservoir layer calculation takes only one clock cycle to complete, enabling the RC system to achieve high throughput even at low operating frequencies. By introducing LUTNet-RC, we designed low-power consumption, under 340 μ W, RC on low-cost FPGAs. We evaluated the proposed systems with practical applications.

II. LUTNET-BASED RESERVOIR COMPUTING DESIGN

A. LUTNet-Based Reservoir Computing

RC is one type of RNN, which comprises of three layers: an input layer, a reservoir layer, and an output layer, as shown in Fig. 1 [2]. The input, reservoir, and output layers have N_i ,

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N_r , and N_o neurons, respectively. W_{ir} , W_{rr} , and W_{ro} represent the weights between the input and reservoir layers, inside the reservoir layer, and between the reservoir and output layers, respectively. Although RC is a simple structure as shown in Fig. 1, its accuracy is comparable to that of conventional DL-based RNNs [5].

LUTNet-RC comprises of a LUT-based reservoir layer and a non-LUT-based output layer [4]. The LUT-based reservoir layer uses LUTNet which is a quantization method focused on FPGA implementation. LUTNet focuses only on the input-output relationship of each neuron and maps it directly to the LUT elements on FPGAs. Therefore, the schematic of the LUT-based reservoir layer is shown in Fig. 1. The LUT-based reservoir neuron is represented by

$$s_i[t] = \begin{cases} -1, & (x_i[t] < 0) \\ 1, & (\text{otherwise}) \end{cases} \quad (1)$$

$$x_i[t] = \sum_{j=1}^{N_r} W_{rri,j} \times s_j[t-1] + \sum_{j=1}^{N_i} W_{iri,j} \times u_j[t], \quad (2)$$

where $s_i[t]$ and $x_i[t]$ are the output state and the internal state of i th reservoir neuron at time t , respectively.

The non-LUT-based output layer is represented by

$$o_k[t] = b_k + \sum_{j=1}^{N_r} s_j[t] \times W_{rok,j} \quad (3)$$

where $o_k[t]$ and b_k are the output value and bias value of k th output neuron, respectively. The non-LUT-based output neuron is implemented as normal multiply-accumulation (MAC) units. Owing to circuit size limitations, it is generally difficult to implement the all MAC units in parallel. One of the techniques that is often used is time-division calculation. The simple time-division calculation (Simple-TDC) scheme implements only one MAC unit per output neuron, which is used repeatedly to realize the calculation in (3).

A key feature of LUTNet-RC is that the LUT-based reservoir layer calculation completes in a single clock cycle. In RC, the reservoir layer accounts for a large part of the computational volume, and in conventional works [2], [3], reservoir layer circuits were operated at a high operating frequency to achieve high throughput of RC system, resulting in high dynamic power consumption. On the other hand, in the case of LUTNet-RC, the reservoir layer calculation is completed in one clock cycle because of the LUT-based reservoir layer. As a result, LUTNet-RC achieves high throughput at low operating frequencies while reducing dynamic power consumption. Furthermore, LUTNet-RC is significantly more efficient in terms of circuit area than previous works [4]. Therefore, LUTNet-RC is suitable for implementing ultralow power consumption RC systems on FPGAs.

B. Low-Cost FPGAs

FPGAs have attracted significant attention as implementation platforms for TinyML [2]. FPGAs enable energy-efficient computation owing to their high parallel processing capabilities and ability to configure circuits optimized for specific applications. However, many commercial FPGAs have high

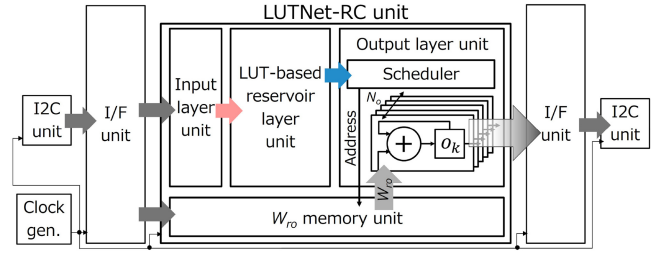


Fig. 2. Proposed LUTNet-RC system for iCE40 series FPGAs.

static power consumption; for example, even relatively low-power FPGAs such as the AMD Xilinx Arty7 reach static power consumption of approximately 100 mW [6]. This value far exceeds the constraint of 340 μ W required for continuous operation for one year on a LR6 battery.

Nonvolatile memory-based FPGAs are also a possible choice for low-power implementations; however, operating clock frequency limitations and manufacturing process technology limit performance and power efficiency in actual TinyML implementations [7].

In this work, we focus on iCE40 series FPGAs from Lattice Semiconductor Inc. The iCE40 series FPGAs are volatile ultralow-power FPGAs, with extremely low static power consumption at the tens of μ W level, making them suitable for TinyML implementation [7], [8]. Although iCE40 series FPGAs have limited circuit resources compared to conventional high-end FPGAs, they are sufficient for implementing LUTNet-RC.

Based on these considerations, we selected the iCE40 series FPGA as our target devices, aiming to implement LUTNet-RC operating below 340 μ W.

C. Proposed System Architecture

In the previous work [4], LUTNet-RC has already been implemented on FPGAs, which is AMD Xilinx KR260 board. However, the system architecture was based on the use of central processing units including operating systems and dynamic random access memories (RAMs), and the power consumption was several W or more.

In this work, we propose an LUTNet-RC system architecture on iCE40 series FPGAs, as shown in Fig. 2. Since iCE40 series FPGAs have inter-integrated circuit (I2C) hard cores, our proposed architecture uses them as interface units. As shown in Fig. 2, the W_{ro} and time-series data from various sensors are input from the left I2C unit, and the right I2C unit outputs the results of calculations based on those input data.

The output layer of LUTNet-RC is implemented using Simple-TDC scheme to reduce circuit size. Therefore, the number of clock cycles required for calculation of the reservoir and output layers are ideally 1 and N_r , respectively.

III. EXPERIMENTS

A. Estimation of Performance

We performed placement and routing (PnR) for the proposed system shown in Fig. 2 on an iCE40 UP5K with iCEcube2 v2020.12 for two configurations $N_r = 256$ and $N_r = 512$

TABLE I
PNR RESULTS OF THE PROPOSED LUTNet-RC SYSTEM

# of reservoir neurons N_r	Max Freq. [MHz]	LUT (%)	Utilization Flip Flops (%)	BRAM (%)
256	33	3625 (68.7)	2130 (40.3)	30 (100)
512	33	4983 (94.4)	2957 (56.0)	30 (100)

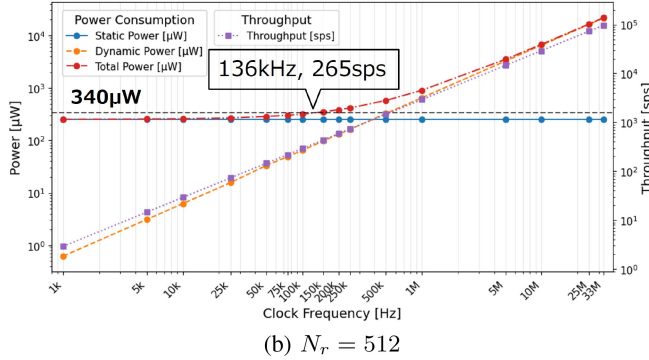
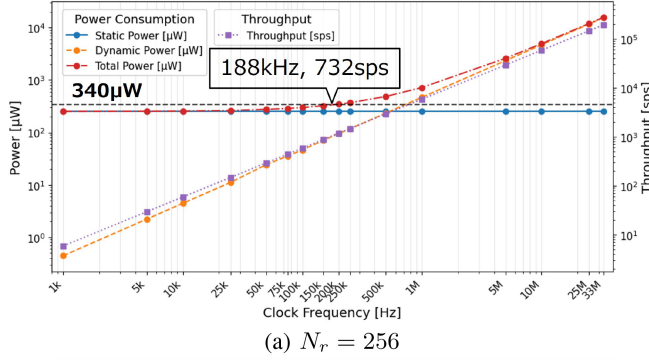


Fig. 3. Power consumption and throughput versus operating clock frequency. (a) $N_r = 256$. (b) $N_r = 512$.

(with $N_i = 16$, $N_o = 8$). Then, based on the PnR results, we estimated the circuit size, power consumption, and throughput (the number of samples per second, sps). The throughput was estimated using the ideal number of clock cycles required for one computation ($1 + N_r$) and the operating frequency.

Table I shows the estimated the circuit size for each model size. Most of the circuit resources on the iCE40 UP5K was used when implementing LUTNet-RC with 512 reservoir neurons. The block RAM (BRAM) usage was 100% in both cases when the number of reservoir neurons was 256 and 512. This is a result of optimization by electrical design automation tools to improve performance. In reality, it is possible to implement the system using much less BRAMs. Actually, if N_r is 512 and W_{ro} is 16-bit precision, 65 kb of memory space is required, which corresponds to about 54% of the BRAM capacity of an iCE40 UP5K.

Fig. 3 shows the power consumption and throughput at various clock frequencies for each model size. LUTNet-RC with 256 reservoir neurons consumed 340 μ W and 732 sps at 188 kHz. LUTNet-RC with 512 reservoir neurons consumed 340 μ W and 265 sps at 136 kHz. Both implementations can operate in real-time while achieving the strict power consumption constraint of less than 340 μ W.

TABLE II
EXPERIMENTAL RESULTS OF PRACTICAL TASKS

	RC model	Accuracy [%]			
		HAR	ED	ECG	Average
LUTNet-RC					
$N_r = 256$	✓	90.2	96.3	88.2	91.6
$N_r = 512$	✓	93.1	98.6	92.7	94.8
CBM-RC	✓	85.7	94.9	85.2	88.6
ESN	✓	87.5	90.3	91.2	89.7
RNN [5]		70.0	56.0	92.3	72.8
GRU [5]		93.5	70.4	92.9	88.9

TABLE III
LUTNet-RC OPERATING CLOCK FREQUENCY AND POWER FOR REAL-TIME PROCESSING

	Required throughput [sps]	$N_r = 256$		$N_r = 512$	
		Freq. [kHz]	Power [μ W]	Freq. [kHz]	Power [μ W]
HAR	50	13	252.10 ^a , 6.11 ^b	26	252.10 ^a , 16.88 ^b
ED	16	4	252.10 ^a , 1.88 ^b	8	252.10 ^a , 5.20 ^b
ECG	250	64	252.10 ^a , 30.07 ^b	128	252.10 ^a , 83.13 ^b

^a: Static Power, ^b: Dynamic Power

B. Practical Applications

We evaluated the proposed system with three practical applications. The first application is human activity recognition (HAR) task [5]. The HAR task classifies a person's activity into six classes based on 12-D input sensor data from a smartphone. The input data sampling rate is 50 Hz. The second application is epilepsy detection (ED) task [5]. The ED task classifies a person's condition into four classes based on 3-D input sensor data. The input data sampling rate is 16 Hz. The third application is electrocardiogram (ECG) anomaly detection task [5]. The ECG task classifies a heart condition into five classes based on 1-D input ECG sensor data. The input data sampling rate is 250 Hz. In addition, we estimated the operating frequency that can achieve the throughput required by each application based on the results in Fig. 3.

Table II shows the accuracy in each task for the proposed LUTNet-RC as well as the typical RC models echo state networks (ESN) and chaotic Boltzmann machine RC (CBM-RC), and the typical DL-based-RNN models RNN and gated recurrent units (GRUs). Table III shows the throughput required for each application, the operating frequency of LUTNet-RC to achieve the requirement, and its power consumption.

From Table II, LUTNet-RC achieves high accuracies in all tasks, and is the best on average. The accuracy outperforms RC models and DL-based RNNs. From Table III, LUTNet-RC meets the throughput requirements of the application even at low operating frequencies and achieves ultra low dynamic power consumption. In all cases, proposed LUTNet-RC system meets the strict power consumption constraint of less than 340 μ W, and can operate for one year on a LR6 battery.

C. Comparison With Previous Works

Table IV shows a comparison between our proposed systems and previous TinyML for time-series data processing on low-cost devices. Our proposed LUTNet-RC system achieves practical throughput of 732 sps and 265 sps with

TABLE IV
COMPARISON WITH PREVIOUS TINYML FOR TIME-SERIES DATA PROCESSING ON LOW-COST DEVICES

	Application	Device	Power [μ W]			Runtime with one LR6 battery [days]
			Static	Dynamic	Total	
LUTNet-RC $N_r = 256$	General^a	iCE40 UP5K	252.10	87.88	339.98	365 ($\times 1.0000$)
	Best Performance ^b			15.54×10^3	15.79×10^3	7.92 ($\times 0.0216$)
	General^c			87.86	339.96	365 ($\times 1.0000$)
	Best Performance ^d			21.47×10^3	21.72×10^3	5.76 ($\times 0.0158$)
[9] (2020)	HAR	Artix-7	-	-	241×10^3	0.52 ($\times 0.0014$)
[6] (2024)	ED	Artix-7	74.00×10^3	0.12	74.12×10^3	1.69 ($\times 0.0046$)
[10] (2019)	ECG	Corex A53*	-	-	900×10^3	0.14 ($\times 0.0004$)
[7] (2019)	Human Detection	iCE40 UP5K	277.00	5.97×10^3	6.24×10^3	20.03 ($\times 0.0549$)
[8] (2024)	Anomaly Detection	iCE40 UP5K	-	-	17×10^3	7.35 ($\times 0.0201$)

Device: default is FPGA, * is MCU. ^a: Freq. is 188 kHz. Throughput is 732 sps. ^b: Freq. is 33 MHz. Throughput is 128,405 sps. ^c: Freq. is 136 kHz. Throughput is 265 sps. ^d: Freq. is 33 MHz. Throughput is 64,327 sps.

N_r of 256 and 512, respectively, while operating with low power consumption. Previous works implemented on the same platform as our system also consume several mW [7], [8].

This significant difference in power consumption is caused by the reduction of dynamic power consumption based on the low operating frequency of our proposed systems. In fact, the Cortex A53 for ECG operates at several MHz [10]. Additionally, while the iCE40 UP5K for human detection [7] has nearly identical static power consumption to our system, its dynamic power consumption is high due to operating at several tens of MHz. The high operating frequencies in these previous works are set to achieve the throughput requirements of several tens of sps for practical applications. In contrast, LUTNet-RC can complete reservoir layer computations, which constitute the majority of the computational load, in a single clock cycle through the LUTNet, enabling practical throughput achievement even at low operating frequencies. As a result, the power efficiency of our proposed system is approximately 10-1000 times superior compared to previous works.

The proposed LUTNet-RC has generality that can be applied to various applications by using RC models. For tasks requiring higher performance, although power consumption increases, the proposed system can operate at up to 33 MHz, achieving throughput of 128,405 sps and 64,327 sps with N_r of 256 and 512, respectively. From Table I, while the circuit scale nearly exhausts the resources with 512 reservoir neurons, it is sufficient to solve practical applications. Furthermore, it is possible to accommodate more reservoir neurons by introducing time-division calculation to reservoir layer and adjustment of hyperparameters related to reservoir layer connections.

In actual systems, sensor and communication peripheral circuits are added to these power consumptions. However, for simple TinyML applications such as anomaly notification, the power consumption increase due to the addition of these peripheral circuits is limited. The proposed LUTNet-RC system on iCE40 UP5K is a promising implementation method for TinyML in battery-powered embedded systems for long-term operation in the healthcare and anomaly detection fields.

IV. CONCLUSION

In this work, we proposed an LUTNet-RC system on iCE40 UP5K to realize ultralow power consumption and low manufacturing cost TinyML. The proposed system can operate at below 340μ W, which enables one-year operation on a LR6 battery. Over 92% accuracy was achieved in HAR, ED and ECG tasks, demonstrating about 10-1000 times better power efficiency compared to previous works. The low-cost FPGA implementation of LUTNet-RC proposed in this work and the achieved low power consumption are promising methods for TinyML, which requires battery-driven long-term operation.

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